



Interior C^2 estimates for a class of sum Hessian equations ☆

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Abstract

In this paper, we mainly study the interior C^2 estimates for a class of sum Hessian equations. We establish the interior estimates and the Pogorelov type estimates for $0 < k < n$. If $k = n$, we derive a weaker Pogorelov type estimates.

Introduction

In this paper, we mainly study the interior C^2 estimates for the following sum Hessian equation $\sigma_k(\eta) + \alpha\sigma_{k-1}(\eta) = f(x, u, Du)$, where u is an unknown function defined on Ω , f is a positive function, $\alpha \geq 0$ and $0 < k \leq n$. Let $\sigma_k(D^2u) = \sigma_k(\lambda(D^2u))$ denote the k -th elementary symmetric function of the eigenvalues of the Hessian matrix D^2u . Namely, for $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, $\sigma_k(\lambda) = \sum_{1 \leq i_1 < \dots < i_k \leq n} \lambda_{i_1} \cdots \lambda_{i_k}$. And define that $\eta = (\eta_1, \eta_2, \dots, \eta_n)$ with $\eta_i = \sum_{j \neq i} \lambda_j$, then η are the eigenvalues of the matrix $(\Delta u)I - D^2u$.

The operator $\sigma_k(\eta)$ appears in many literatures. The origin of the operator comes from the Gauduchon conjecture [14], [32] in complex geometry. Harvey-Lawson [19], [20] introduced and investigated the $(n-1)$ -plurisubharmonic functions, which satisfy the nonnegative definite complex matrix $(\Delta u)I - D^2u$, and solved the Dirichlet problem with $f = 0$ on suitable domains. And in the work of Fu-Wang-Wu [12], the operator can be regarded as a form-type Calabi-Yau equation, and Tosatti-Weinkove [33] solved that on Kähler manifolds.

The equation $\sigma_k(\eta) = f$ is concerned with p -convex hypersurface in prescribed curvature problem. Let M be a hypersurface in $\mathbb{R}^{n+1}$. For given $1 \leq p \leq n$, call M p -convex if $\kappa(X)$ satisfy $\kappa_{i_1} + \cdots + \kappa_{i_p} \geq 0$ for each $X \in M$, where $\kappa(X) = (\kappa_1, \dots, \kappa_n)$ are the principal curvatures of M . For prescribed curvature problem $\sigma_k(\eta) = f$ in $\mathbb{R}^{n+1}$, Chu-Jiao [4] investigated the existence of hypersurfaces. Dong [10] proved the existence of p -convex hypersurface for $p \geq \frac{n}{2}$ with prescribed curvature. When $1 \leq p \leq n$, the p -convex hypersurface was studied intensively and extensively by Sha [30], [31], Wu [34] and Harvey-Lawson [21].

A lot of variant forms of the equation $\sigma_k(\eta) = f$ have also been investigated, and one of the form is the Hessian quotient equation $\frac{\sigma_k(\eta)}{\sigma_l(\eta)} = f$. For instance, Chen-Tu-Xiang [6] established the Pogorelov type estimates when $l > 0$ and $l + 2 \leq k \leq n$. Dong-Wei [11] proved that the Hessian quotient equation is strictly elliptic in Γ'_{k+1} (see Definition 1 below). And Chen-Dong-Han [3] derived the interior estimates and Pogorelov type estimates for the case $0 \leq l < k < n$.

If η is replaced with $\lambda(D^2u)$, the operator $\sigma_k(\eta)$ will come to the following classic k -Hessian equation $\sigma_k(D^2u) = f(x, u, Du)$. The existence of the solution to the above equation is generally obtained through continuity method and prior estimates. For the right hand side function f not depending on the gradient term, Caffarelli-Nirenberg-Spruck [5] solved the Dirichlet problem. When the right hand side function f depends on the gradient term, how to establish the C^2 estimates of the solution is a long-standing problem. There have been some works related to the C^2 estimates, such as [7], [15], [22], [28], [35].

In recent years, the fully non linear equations derived from linear combinations of primary symmetric functions have received increasing attention. For example, the following special

Lagrangian equation $\text{Imdet}(\delta_{ij} + iu_{ij}) = \sum_{k=0}^{[(n-1)/2]} (-1)^k \sigma_{2k+1}(\lambda(u_{ij})) = 0$ was derived by Harvey and Lawson in their study of the minimal submanifold problem [18]. The fully nonlinear partial differential equations $P_m(u_{ij}) = \sum_{k=0}^{m-1} (l_k^+)^{m-k}(x) \sigma_k(u_{ij}) = g^{m-1}(x)$ were studied by Krylov [23] and Dong [8]. The following curvature equations and complex Hessian equations

$\sum_{s=0}^k \alpha_s \sigma_k(\kappa(\chi)) = f(\chi, v(\chi)), \quad \chi \in M, \sum_{s=0}^k \alpha_s \sigma_k(\lambda) = f(z, u, Du)$ were studied by Li-Ren-Wang [25] and Dong [9]. For the following curvature equations

$\sigma_k(W_u(x)) + \alpha \sigma_{k-1}(W_u(x)) = \sum_{l=0}^{k-2} \alpha_l(x) \sigma_l(W_u(x)), \quad x \in \mathbb{S}^n$, Guan-Zhang [17] and Zhou [36]

studied the curvature estimates and the interior gradient estimates respectively. Recently, Liu and Ren [24] discussed the Pogorelov type C^2 estimations for $(k-1)$ -convex and k -convex solutions of the following sum Hessian equations $\sigma_k(\lambda(u_{ij})) + \alpha \sigma_{k-1}(\lambda(u_{ij})) = f(x, u, \nabla u)$.

For a class of sum hessian equations like (1.1), with the inspiration from Chen-Dong-Han [3] and Chen-Tu-Xiang [6], we study the interior estimates and Pogorelov type estimates.

First, we give the definition of k -convex and k -admissible.

Definition 1

For a domain $\Omega \subset \mathbb{R}^n$, a function $u \in C^2(\Omega)$ is called k -convex if the eigenvalues $\lambda(x) = (\lambda_1(x), \dots, \lambda_n(x))$ of the Hessian $\nabla^2 u(x)$ are in Γ_k for all $x \in \Omega$, where Γ_k is the Garding's cone $\Gamma_k = \{\lambda \in \mathbb{R}^n | \sigma_m(\lambda) > 0, m = 1, \dots, k\}$. Similarly, a function $u \in C^2(\Omega)$ is called k -admissible if the eigenvalues $\lambda(x) = (\lambda_1(x), \dots, \lambda_n(x))$ of the Hessian $\nabla^2 u(x)$ are in Γ'_k for all $x \in \Omega$, where $\Gamma'_k = \{\lambda \in \mathbb{R}^n | \sigma_m(\eta) > 0, m = 1, \dots, k\}$,

Here we list the main results.

Theorem 2

Suppose that $u \in C^4(B_R(0))$ is a $(k-1)$ -admissible solution of the sum Hessian equation (1.1), where $B_R(0)$ is a ball centered at the origin with radius R in $\mathbb{R}^n$, $0 < k < n$ and $f \in C^2(B_R(0) \times \mathbb{R} \times \mathbb{R}^n)$ with $0 < m \leq f \leq M$. Then $|D^2 u(0)| \leq C(1 + \frac{\sup |Du|}{R})$, where C is a positive constant depending only on $n, m, M, R \sup |Df|$ and $R^2 \sup |D^2 f|$.

Following the argument of the proof of Theorem 2, we can also obtain the Pogorelov type estimates

$$\text{for the Dirichlet problem} \begin{cases} \sigma_k(\eta) + \alpha \sigma_{k-1}(\eta) = f(x, u, Du), & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Theorem 3

Suppose that $u \in C^4(\Omega) \cap C^2(\overline{\Omega})$ is a $(k-1)$ -admissible solution of the problem (1.3) in a bounded domain $\Omega \subset \mathbb{R}^n$, where $0 < k < n$ and $f \in C^2(\overline{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ with $f > 0$. Then $(-u)|D^2 u| \leq C$, where C depends only on $n, k, |f|_{C^2}$ and $|u|_{C^1}$.

For the case $k = n$, we still have the following result.

Theorem 4

Suppose that $u \in C^4(\Omega) \cap C^2(\overline{\Omega})$ is a $(n-1)$ -admissible solution of the problem (1.3) in a bounded domain $\Omega \subset \mathbb{R}^n$, where $k = n$ and $f \in C^2(\overline{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ with $f > 0$. Then $(-u)^\beta |D^2 u| \leq C$, where C depends only on $n, |f|_{C^2}$ and $|u|_{C^1}$.

The organization of our paper is as follows. In Section 2, we give the preliminary knowledge. The proofs of Theorem 2 and Theorem 4 are given respectively in Section 3 and Section 4.

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Section snippets

Preliminary

We recall some basic definition and properties of elementary symmetric function as follows.

For $1 \leq k \leq n$, the k^{th} elementary symmetric function is defined as

$$\sigma_k(\lambda) = \sum_{1 \leq i_1 < \dots < i_k \leq n} \lambda_{i_1} \cdots \lambda_{i_k} \text{ for } \lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n \text{ and } 1 \leq k \leq n. \text{ For convenience,}$$

we further set $\sigma_0(\lambda) = 1$ and $\sigma_k(\lambda) = 0$ for $k > n$ or $k < 0$.

We define the sum Hessian function as $S_k(\lambda) = \sigma_k(\lambda) + \alpha \sigma_{k-1}(\lambda)$ for $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{R}^n$ and $1 \leq k \leq n$.

Now we will list some algebraic identities and properties of σ_k and S_k . Denote

$$(\lambda|a) = (\lambda_1, \dots, \lambda_{a-1}, \lambda_{a+1}, \dots, \lambda_n).$$

Proposition 5

We have

$$(1) S_k^{pp}(\lambda) := \frac{\partial S_k}{\partial \lambda_k} \dots$$

...

Proof of Theorem 2

In this section, we prove the interior a priori Hessian estimate for (1.1). We will use the method similar to [3].

Assume $R = 1$, otherwise we consider $\tilde{u}(x) := \frac{1}{R^2} u(Rx)$ and the equation $S_k(\tilde{\eta}) = f(Rx)$ in $B_1(0)$. Let $\tau(x) = \tau(D^2 u(x)) = (\tau_1, \dots, \tau_n) \in \mathbb{S}^{n-1}$ be the continuous eigenvector field of $D^2 u(x)$ corresponding to the maximum eigenvalue, and we consider the auxiliary function

$$\phi(x) = \rho(x) g\left(\frac{|Du|^2}{2}\right) u_{\tau\tau}, \text{ where } \rho(x) = (1 - |x|^2), \text{ and } g(t) = \left(1 - \frac{t}{A}\right)^{-1/3} \text{ with } A = \sup |Du|^2 > 0. \text{ Thus } g'(t) = \frac{1}{3A} \left(1 - \frac{t}{A}\right)^{-4/3} \text{ and } g''(t) = \frac{4}{9A^2} \left(1 - \frac{t}{A}\right)^{-7/3} \dots$$

Proof of Theorem 4

In this section, we will use the ideas in [6] to prove Theorem 4. For convenience, we suppose

$U[u] = (\Delta u)I - D^2 u$ and introduce the following notations.

$$F(U) = S_n^{\frac{1}{n}}(U), \quad T(D^2 u) = F(U), \quad F^{ij} = \frac{\partial F}{\partial U_{ij}}, \quad F^{ij,rs} = \frac{\partial^2 F}{\partial U_{ij} \partial U_{rs}}, \text{ From the notation we have}$$

$$T^{ii} = \frac{\partial T}{\partial u_{ii}} = \sum_{p,q} \frac{\partial F}{\partial U_{pq}} \frac{\partial U_{pq}}{\partial u_{ii}} = \sum_{j=1}^n \frac{\partial F}{\partial U_{jj}} \frac{\partial U_{jj}}{\partial u_{ii}} = \sum_{j=1}^n F^{jj} - F^{ii}. \text{ where we used}$$

$U_{ii} = \eta_i = \sum_{j=1}^n u_{jj} - u_{ii}$ in the last equation. Suppose $u \in C^4(\Omega) \cap C^2(\overline{\Omega})$ is a solution of equation (1.1), $\eta \in \tilde{\Gamma}_n$. Without loss of generality, by the maximum principle we assume $u < 0$ in Ω . Consider the ...

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